



भारतीय संगणक एवं प्रौद्योगिकी संस्थान

INDIAN INSTITUTE OF COMPUTING AND TECHNOLOGY

AFFILIATED: I-STEM, OFFICE OF THE PRINCIPAL SCIENTIFIC ADVISER TO THE GOVERNMENT OF INDIA

Project -1 Summer Internship Program in AI&ML Machine Learning 2026.

AI- Powered Fake News Detection Using Text Classification

Problem Statement

Build a machine learning pipeline **from scratch** to classify news articles as *real* or *fake*. Students must implement preprocessing, feature extraction, model training, and evaluation without relying on pre- built solutions.

Data Sources

Kaggle: Fake News Detection Dataset ([kaggle.com in Bing](https://www.kaggle.com/datasets/pranavkumar1997/fake-news-detection-dataset))

UCI Repository: Fake News Dataset

Optional: Use *NewsAPI* to scrape recent articles and manually label a subset.

30- Day Workflow

Week	Tasks
Week 1	Collect dataset, clean text (remove stopwords, punctuation), tokenize manually
Week 2	Implement Bag- of- Words, TF- IDF, embeddings; exploratory data analysis
Week 3	Build models (KNN, Logistic Regression, Random Forest, Neural Net)
Week 4	Evaluate models, visualize results, prepare report & presentation

Algorithms to Implement

KNN (Non- Parametric)

Logistic Regression (Parametric)

Random Forest (Ensemble)

Simple Neural Network (Deep Learning)



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Python Code Skeleton

```
python
# Week 1: Data Loading & Cleaning
import pandas as pd
import re
from sklearn.model_selection import train_test_split

data = pd.read_csv("train.csv") # Kaggle dataset
X = data['text']
y = data['label']

def clean_text(text):
    text = re.sub(r'\W', ' ', text) # remove punctuation
    text = text.lower()
    return text

X = X.apply(clean_text)

# Week 2: Feature Engineering
from sklearn.feature_extraction.text import TfidfVectorizer
vectorizer = TfidfVectorizer(max_features=5000)
X_vec = vectorizer.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(X_vec, y, test_size=0.2,
random_state=42)

# Week 3: Model Building
from sklearn.neighbors import KNeighborsClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.neural_network import MLPClassifier
```



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```
models = {  
    "KNN": KNeighborsClassifier(n_neighbors=5),  
    "LogReg": LogisticRegression(max_iter=1000),  
    "RandomForest": RandomForestClassifier(n_estimators=100),  
    "NeuralNet": MLPClassifier(hidden_layer_sizes=(100,), max_iter=300)
```

```
# Train & Evaluate
```

```
from sklearn.metrics import accuracy_score, classification_report
```

```
for name, model in models.items():  
    model.fit(X_train, y_train)  
    preds = model.predict(X_test)  
    print(f"{name} Accuracy:", accuracy_score(y_test, preds))  
    print(classification_report(y_test, preds))
```

How AI & ML can be Used with proper expansion of the following

AI: Automates the detection of misinformation.

ML: Learns patterns in text (word frequency, context) to classify articles.

NLP: Preprocessing (tokenization, TF- IDF) converts raw text into numerical features.

Evaluation: Accuracy, precision, recall, F1- score, confusion matrix.



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Documentation Guidelines

Students must submit:

1. **Introduction** — Problem statement, importance of fake news detection.
2. **Dataset Description** — Source, size, features.
3. **Methodology** — Preprocessing, feature extraction, algorithms.
4. **Results** — Accuracy, precision, recall, F1- score, confusion matrices.
5. **Discussion** — Compare parametric vs. non- parametric models.
6. **Conclusion** — Insights, limitations, future scope.
7. **Appendix** — Python code, test data samples.
8. **Format & Standard**: IEEE